2 Calibration of the model

2.1 Exponents of the Composite Weibull distribution

The shape of the Composite Weibull distribution is determined by the shape parameters k_1, k_2, and $\tilde{H}_{tr}$. In this section assumptions about the value of the exponents k_1 and k_2 are presented.

The value for k_1 is based on initial inspection of measured wave height distributions of the calibration data. Measured wave height distributions plotted on Rayleigh scale show a straight line for the lower wave heights. Therefore it is assumed that the first part of the wave height distribution, $F_1(H)$, is Rayleigh shaped. This means that the exponent k_1 equals the exponent of the Rayleigh distribution ($k_1 = 2$).

For further analysis use is made of the parameter:

$$\psi = \frac{\sqrt{m_0}}{d} \tag{2.1}$$

in which m_0 is the variance of the water surface elevation, which equals the zero-th spectral moment and d denotes the water depth. The properties of the normalised wave height distribution are assumed to be characterised mainly by a relative local wave height, such as H_s/d. Here, instead of H_s/d the degree of saturation ψ is used to characterise the wave deformation process on shallow foreshores.

Initial visual fitting of the Composite Weibull distribution to wave height distributions measured on foreshores with different slopes yielded a relation between exponent k_2 and the degree of saturation as presented in Figure 5.

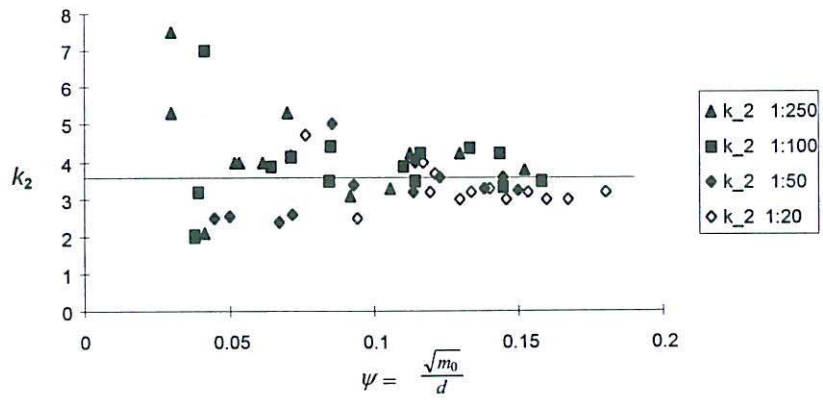

Figure 5: Exponent k_2 versus degree of saturation, for different foreshore slopes

Figure 5 shows a decrease of scatter in k_2 as the degree of saturation increases. High scatter for low ψ-values is understandable: When the Composite Weibull distribution is fitted to measured wave height distribution of wave fields with low degrees of saturation the distribution hardly deviates from the Rayleigh distribution. This results in fitting of the

second part of the Composite Weibull distribution to only a few measured wave heights with a low probability of exceedance. This explains the scatter. At the same time, it implies that the value of k_2 is not important for low ψ-values. Since the higher degrees of saturation represent the area of interest, Figure 5 supports an assumption of a constant exponent k_2. A least squares optimisation method yields k_2=3.6.

Still, the average exponent k_2 on shallow foreshores (slope 1:20 in Figure 5) is slightly smaller than 3.6 and on milder foreshores k_2 is slightly higher than 3.6. So one could try to obtain a function for k_2 which decreases with increasing foreshore slope. Here, for all slopes k_2=3.6 is used.

2.2 Transitional wave height

The transitional wave height is the wave height at which the wave height distribution changes its shape. Due to the assumption of two constant exponents the shape of the Composite Weibull distribution totally depends on the normalised transitional wave height $\tilde{H}_{tr}$: Equation 1.7 together with the normalised continuity condition form a set of two (implicit) algebraic equations with three independent parameters. By assuming k_1=2 and k_2=3.6 the Composite Weibull distribution is related to the only shape parameter left, $\tilde{H}_{tr}$. Figure 6 shows how all characteristic normalised wave heights are related to $\tilde{H}_{tr}$ (for example, x=1/3 yields $\tilde{H}_{1/3}$).

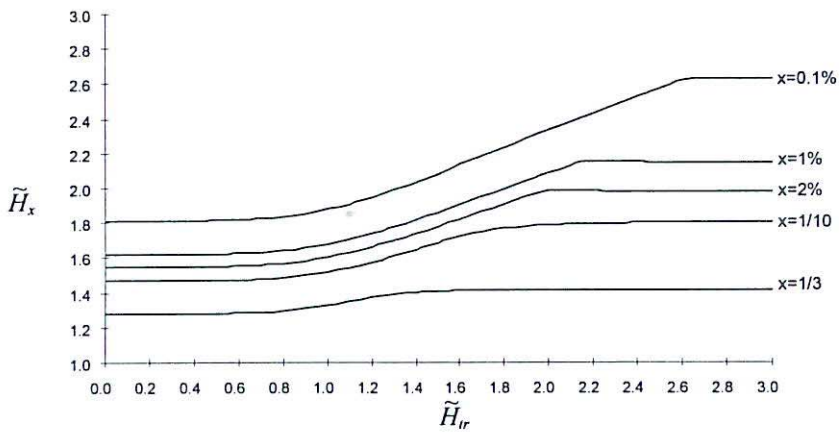

Figure 6: Characteristic nondimensional wave heights related to the $\tilde{H}_{tr}$

In Table B.1 the relations, as shown in Figure 6, are presented in tabular form. To use the preceding results one still needs to determine $\tilde{H}_{tr}$ for given depth, foreshore slope and energy spectrum (m_0, T_p, etc.). In the following sections H_{tr} is related to the local water depth, foreshore slope and wave period. Then, in order to scale the wave height distribution to the wave field of concern, H_{rms} is related to the total wave energy.

2.2.1 Depth and foreshore slope

The transitional wave height is the wave height at which the wave height distribution abruptly changes its shape. The change in shape is a result of depth-induced breaking of the waves exceeding the transitional wave height. Therefore the following hypothesis is posed:

$$H_{tr} \sim d \tag{2.2}$$